Using StoryMaps as a Tool for Science Communication: a Case Study with the Wyandotte Nation and Flood Risk

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INTRODUCTION

Flooding presents significant risks to communities, particularly those in regions with complex governance structures like Ottawa county, which includes nine distinct tribal governments. Effective communication of flood risk data is crucial for building community resilience and ensuring preparedness. This study investigates the value of ArcGIS StoryMaps in bridging communication gaps and enhancing resilience to flood-prone tribal communities.

By blending traditional storytelling with modern research, we aim to create an engaging and comprehensive platform for science communication. This approach facilitates better understanding and collaboration among diverse audiences. Additionally, integrating various media forms such as maps, images, and narrative text makes the information more accessible and relatable, further supporting community efforts to mitigate flood risks.

METHODS

- **Partnerships with Indigenous Partners** - Developed jointly with the Wyandotte Nation, this study emphasizes collaboration and mutual respect
- **Selection of ArcGIS Storymaps** ArcGIS StoryMaps were chosen for their ability to integrate diverse media and present information interactively. It has potential to bridge cultural and communication gaps, making complex data accessible

Community Walkthrough

The StoryMap was developed after physically experiencing the community with local partners. Walking through the community allowed us to gather firsthand insights and understand the local context better, ensuring that the StoryMap was grounded in the lived experiences of the residents.

Storytelling Element

To resonate with our partners, we used the narrative of a man rowing down the river in a canoe. This storytelling element helped organize the StoryMap and tell a more effective story by blending traditional storytelling and modern research perspectives.

Data Collection and Analysis

Data was collected from various sources, including FEMA floodplain maps, local historical data on chat piles, high resolution aerial photography, and first-hand accounts of flood events.

Stakeholder Engagement

The StoryMap was used to engage stakeholders, including tribal community members and government officials. Feedback from these sessions was incorporated to refine and ensure it addresses the community's needs and concerns.

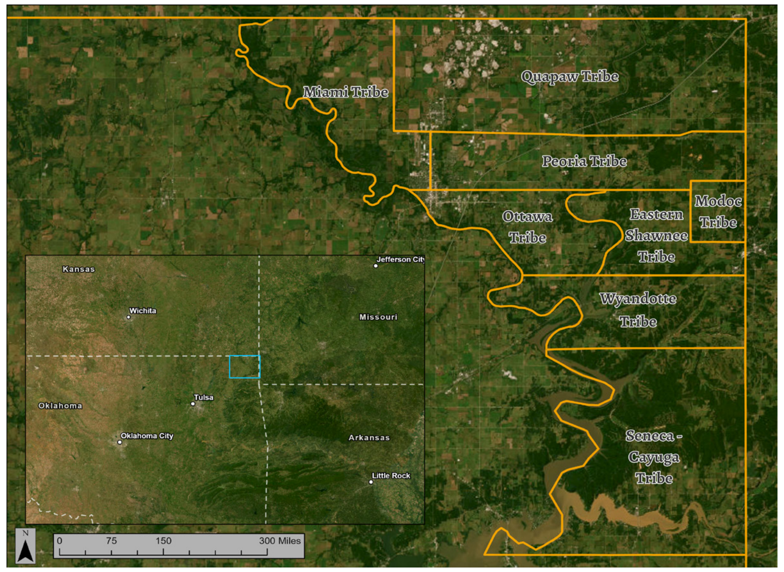

Figure 1. Ottawa County is the densest concentration of sovereign nations anywhere in the world. With a total of 9 tribes, each has a distinct culture, perception, government, and resources.

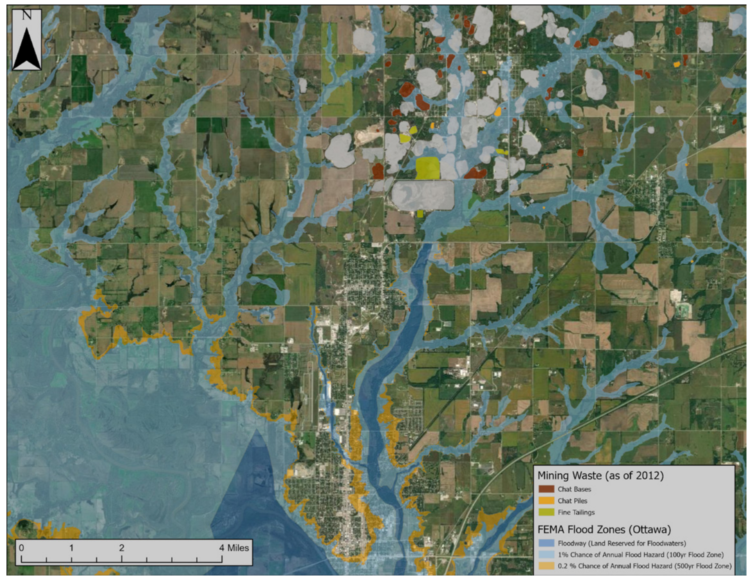

Figure 2. This map shows documented mining waste piles along with the FEMA floodzone near Miami, OK. During flood events, toxic minerals are transported into the county seat and largest urban area in the county.

Figure 4. The proposed location of a water gauge in Seneca, Missouri to give Wyandotte citizens and residents vital time to evacuate. and prepare.

PRELIMINARY FINDINGS

A water gauge in Seneca, MO: Seneca routinely floods around an hour to an hour and a half before Wyandotte, however warning occurs by visual observation of local residents rather than a monitored gauge.

Engineering with Nature: Slowing water upstream can reduce the impact at the trestle and reduce further harmful sediment transport. This upstream river control could be accomplished with Engineering with Nature principles that could filter heavy metals and naturally slow the water.

Water Diversion: Water diversion structures could reduce the impact that the trestle has on the flooding of Wyandotte. Engineering with nature could be used to trap some of the sediment and metals at these points, and reduce the amount that make it downstream.

The Social Component North Oklahoma is a very politically complicated region; home to 9 federally recognized tribes, 10 cities, and a mess of federal jurisdictional boundaries (such as EPA and USCAE zones. Additionally, there are state borders on two sides, therefore, improving communication between these zones will lead to better problem solving.

Figure 5. Quapaw Tribe Fire and Emergency Medical Service have Memorandums Of Understanding with 4 cities and 4 tribes in Ottawa County. As such their swift water rescue is one of the best in the nation, and this image shows them responding to Hurricane Florence in 2018.

Figure 6. An overhead look at the size of chat piles in Picher, that measured over 200 feet high at their peak.

Figure 7. Aerial footage of Miami, OK during the most recent major flooding event in 2019.

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DISCUSSION AND NEXT STEPS

The use of ArcGIS StoryMaps in this project has been an effective tool for science communication. By integrating maps, images, and narratives, StoryMaps create an engaging platform that enhances understanding among diverse audiences. Feedback from community and government officials highlighted the tool's ability to present complex flood risk data accessibly, promoting better preparedness and resilience.

Future work will focus on deepening community engagement through focus groups and listening sessions at the Intertribal Council as part of the NOAA project on flood risk perception. We will use this data to improve existing models and further support community decision-making and preparedness.